

Neuro-Symbolic AI in Indian Education



For Prelims: Large Language Models , Neuro-Symbolic AI, Neural networks , Deep learning , National Education Policy (NEP) 2020 , DIKSHA

For Mains: Artificial Intelligence in India's education sector, Neuro-Symbolic AI and explainable AI systems, Challenges of AI implementation in Indian classrooms, NEP 2020 and technology-enabled learning

Source: TH

Why in News?

Technology experts and educational researchers have highlighted the potential of **Neuro-Symbolic Artificial Intelligence (NSAI)** as a more suitable, reliable, and effective framework for the Indian education system compared to traditional generative models like **Large Language Models (LLMs)**.

Summary

- **Neuro-Symbolic AI (NSAI)** combines neural networks with symbolic reasoning to create explainable, multilingual, and fact-based AI systems that align with National Education Policy goals of conceptual and inclusive learning.
- NSAI can reduce AI hallucinations, bridge vernacular barriers, and support rural education through lightweight offline deployment, but challenges such as digital divide, teacher training, linguistic diversity, and data privacy must be addressed for effective implementation in India.

What is Neuro-Symbolic AI?

- **Hybrid Architecture:** Unlike standard LLMs that rely entirely on statistical next-word prediction, **NSAI** fuses data-driven **neural networks** with rule-based symbolic logic systems .
- **The Neural Component (Perception):** Utilizes **deep learning** (neural networks) for pattern recognition.
 - It excels at processing unstructured, "messy" data such as recognizing a student's handwriting,

analyzing a spoken query in a regional language, or extracting features from an image.

- **The Symbolic Component (Reasoning):** Relies on explicit, human-readable logic rules, constraints, and **Knowledge Graphs / Ontologies** .
- **The Synergy:** In an NSAI system, the neural network acts as the **"eyes and ears,"** converting unstructured data into symbols, while the symbolic engine acts as the **"logical brain,"** applying strict rules to those symbols to generate a verifiable, fact-based answer.
 - This integration helps AI become more accurate, explainable, and human-like in decision-making.
- **Importance of Neuro-Symbolic AI:** The technology **improves transparency, trustworthiness, and explainability** in AI systems.
 - It is considered important for developing **human-centric AI capable of handling complex real-world problems responsibly**.
- **Applications:** NSAI has applications in healthcare, robotics, autonomous vehicles, cybersecurity, scientific research, and advanced decision-support systems where both pattern recognition and logical reasoning are essential.

What are the Limitations of Traditional LLMs in the Indian Classroom?

- **Infrastructure and Energy Mismatch:** Frontier models like **GPT-4** require massive data centers with staggering energy footprints.
 - This is a severe mismatch for India, where **only 47% of rural schools possess functional computers** , and high-bandwidth internet remains severely constrained, widening the digital divide rather than democratizing education.
- **The Vernacular Data Trap:** Standard models are heavily English-dominant. When evaluated against **India's 22 constitutionally recognized languages and regional dialects** , purely statistical models suffer from **low contextual data availability**.
 - They often resort to clunky translations that distort conceptual accuracy.
- **Cultural Disconnect:** Western-trained models risk **embedding foreign cultural biases or missing critical local context** , making them unreliable for culturally rooted pedagogy.
- **"Hallucination" Crisis:** When pushed beyond its training data, an LLM will **confidently invent historical dates, fabricate scientific formulas,** or cite non-existent academic papers.
 - A student or an overworked teacher relying on the AI may not possess the immediate subject matter expertise to spot these errors. This creates a **"hallucination crisis" that corrupts the learning process and actively teaches incorrect information**.
- **Amplification of Rote Learning:** Because traditional generative AI acts as a **statistical pattern-matcher rather than a logical processor**, it gives answers without encouraging **conceptual clarity**, failing to stimulate original scientific reasoning or align with the core goals of the **National Education Policy (NEP) 2020** .
- **The "Black Box" Problem:** If an LLM gives a student the wrong answer to a calculus problem, it cannot explain *why* it made the error. This lack of interpretability prevents teachers from tracing the student's learning gap.

What is the Strategic Significance of NSAI for Indian Education?

The integration of NSAI aligns perfectly with the mandates of the **National Education Policy (NEP) 2020** , which emphasizes conceptual clarity, multilingualism, and reducing cognitive load.

- **Factual Grounding via Curricular Ontologies:** Unlike LLMs, NSAI can be hardcoded with structured knowledge bases, such as mapping the entire **NCERT curriculum** into a logic tree.
 - The AI's outputs are strictly constrained by these facts. If a student asks a physics question, the neural network understands the query, but the symbolic engine retrieves the answer from the **verified NCERT ontology, eliminating hallucinations.**
- **Overcoming the Vernacular Barrier:** NSAI bypasses this by combining neural translation with **explicit symbolic grammatical rules** of Indian languages (e.g., Paninian grammar logic for Sanskrit or Hindi).
 - By hardcoding the grammatical rules, the system requires exponentially less training data to achieve high accuracy in regional languages, aiding initiatives like **Bhashini** .
- **Explainable "Knowledge Tracing":** In a **high Pupil-Teacher Ratio (PTR)** environment, teachers cannot manually analyze every student's mistakes.
 - NSAI performs **"Knowledge Tracing."** If a student fails a multi-step algebraic equation, the **symbolic engine breaks down the exact logic pathway the student used**, pinpointing the specific micro-concept they misunderstood (e.g., applying the distributive property incorrectly).
 - The AI component provides step-by-step explainable feedback, acting as a highly accurate tutor rather than an answer generator.
- **Frugal Innovation and Compute Efficiency:** Training massive LLMs requires prohibitive computational resources (like **GPUs**) and power.
 - Because NSAI relies on pre-programmed logic for reasoning rather than learning it from scratch via brute-force data consumption, these models are significantly **smaller, lighter, and cheaper to run.**
 - Built on lightweight principles like the **C3AN framework** (AI systems that are reliable, cost-effective, and safe for high-stakes enterprise applications) , **they can be downloaded entirely onto low-cost, second-hand smartphones and run on low internet bandwidth** , bypassing rural connectivity deficits.

Indigenous Case Studies

Indian institutions are already successfully piloting this logic-first architecture:

- **Project PrahelikaAI (IIT Kharagpur):** Uses logic puzzles to stimulate cognitive reasoning. It features a 24/7 digital tutor that tracks student delays and offers localized hints in Hindi and Bengali, building a personalized learning profile of recurring misconceptions.
- **C3AN Framework & Edge Deployment:** Designed specifically for rural deployment, enabling a student in rural Odisha to master complex engineering concepts natively in Odia on a low-end device, completely offline.

What are the Challenges in Implementation of NSAI in India's Education System?

- **Knowledge Engineering Bottleneck:** The symbolic component of NSAI depends on highly detailed

knowledge graphs and ontologies that must be manually designed by experts.

- Digitising India's vast and multi-layered curriculum across NCERT, State Boards, and technical education systems into machine-readable logic structures is **extremely complex, time-consuming, and resource-intensive**.
- **Linguistic and Dialectal Diversity:** India's linguistic diversity creates major challenges for NSAI deployment.
 - Building accurate **symbolic reasoning systems for multiple regional languages** and dialects requires extensive linguistic datasets and localized logic frameworks, which remain **underdeveloped for many Indian languages**.
- **Digital Divide and Infrastructure Gaps:** Large sections of rural India still face inadequate electricity, internet access, and digital infrastructure.
 - Deploying NSAI across **fragmented hardware ecosystems**, including outdated smartphones and poorly equipped schools, creates serious **operational and software optimization challenges**.
- **Socio-Emotional Blindspot:** NSAI can identify academic weaknesses but cannot fully understand **emotional, psychological, or socioeconomic factors** affecting student performance.
 - Issues such as **malnutrition, stress, family environment, mental health, and lack of motivation remain beyond the data-processing** capability of AI systems.
- **Risk of Over-Reliance on Algorithms:** AI-based educational diagnostics may reduce the importance of **human mentorship, empathy, counseling, and classroom interaction**.
 - Education is **not only cognitive but also deeply social and emotional**, which AI systems cannot fully replicate.
- **Equity and Inclusion Concerns:** If implemented unevenly, NSAI could widen educational inequality between urban and rural students, private and government schools, and digitally connected versus **digitally excluded populations**, thereby undermining inclusive education goals under the **National Education Policy**.

What Measures are Needed to Make NSAI Inclusive and Effective in India's Education System?

- **Integration with Digital Public Infrastructure (DPI):** NSAI-powered tutoring systems can be integrated into existing educational DPIs like **DIKSHA (Digital Infrastructure for Knowledge Sharing)** to democratize access to high-quality, personalized learning for students in remote areas.
- **Building a "Bharat-Ontology"(Open-Source Knowledge Graphs):** Under the **IndiaAI Mission**, the Ministry of Education must collaborate with academic institutions (IITs/IITs) and ed-tech firms to build open-source, curriculum-aligned knowledge graphs.
- **Frugal Engineering and Edge Deployment:** To address India's digital divide, NSAI systems must be lightweight, offline-friendly, and compatible with low-cost devices.
 - Edge-based deployment enables students in rural areas with poor internet and electricity access to use real-time AI tutoring directly on smartphones or tablets without continuous connectivity.
- **Building NISTHA 2.0:** Teacher training programs like **National Initiative for School Heads' and Teachers' Holistic Advancement** must be upgraded to help educators interpret NSAI-generated learning diagnostics and tailor lesson plans accordingly.
 - NSAI should also automate repetitive administrative and grading tasks, allowing teachers to focus more on mentoring and students' socio-emotional development.

- **Strict DPDP Act Compliance:** The deployment of NSAI must strictly adhere to the **Digital Personal Data Protection (DPDP) Act, 2023**.
 - Student data must be anonymized, localized, and used exclusively for pedagogical improvement, with strict prohibitions against commercial exploitation by third-party vendors.
- **Bias Auditing:** The symbolic logic and knowledge graphs must be regularly audited by independent pedagogical bodies to ensure they do not inadvertently encode gender, caste, or regional biases into the AI's reasoning pathways.

Conclusion

Aligning with the NEP 2020 and **Sustainable Development Goal 4 (Quality Education)**, NSAI represents a profound **pedagogical equalizer for India**. By fusing neural processing with strict factual logic, it effectively eliminates AI hallucinations and language barriers to democratize learning. This transparent digital tutor empowers rural students and liberates **teachers to focus on true human mentorship**.

Drishti Mains Question:

Discuss the significance of Neuro-Symbolic AI in transforming India's education system. How does it differ from traditional Large Language Models?

Frequently Asked Questions (FAQs)

1. What is Neuro-Symbolic AI (NSAI)?

NSAI is a hybrid AI framework that combines neural networks for pattern recognition with symbolic logic systems for reasoning, making AI more explainable, accurate, and trustworthy.

2. Why is NSAI considered better than traditional LLMs for Indian education?

NSAI reduces hallucinations, supports multilingual learning, provides explainable reasoning, and works efficiently on low-cost devices with limited internet connectivity.

3. How does NSAI support the goals of NEP 2020?

NSAI promotes conceptual learning, multilingual education, reduced cognitive load, and inclusive digital education aligned with the objectives of National Education Policy.

4. What are the major challenges in implementing NSAI in India?

Key challenges include digital divide, linguistic diversity, lack of infrastructure, teacher training deficits, data privacy concerns, and the difficulty of building curriculum-based knowledge graphs.

5. Why is the DPDP Act important for NSAI deployment in education?

The Digital Personal Data Protection Act, 2023 ensures student data privacy, localization, anonymization, and protection against misuse by third-party platforms.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/JvoPzjpKKqU](https://www.youtube.com/embed/JvoPzjpKKqU)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. With the present state of development, Artificial Intelligence can effectively do which of the following? (2020)

1. Bring down electricity consumption in industrial units
2. Create meaningful short stories and songs
3. Disease diagnosis
4. Text-to-Speech Conversion
5. Wireless transmission of electrical energy

Select the correct answer using the code given below:

- (a) 1, 2, 3 and 5 only
- (b) 1, 3 and 4 only
- (c) 2, 4 and 5 only
- (d) 1, 2, 3, 4 and 5

Ans: (b)

Mains

Q. National Education Policy 2020 is in conformity with the Sustainable Development Goal-4 (2030). It intends to restructure and reorient the education system in India. Critically examine the statement. (2020)

RBI's Intervention to Defend the Rupee



Source: IE

Why in News?

The **Indian Rupee (INR)** has faced severe **depreciation**, nearing a historic low of **97 against the US Dollar in May 2026**. To arrest this steep **depreciation**, the **Reserve Bank of India (RBI)** has deployed a multi-pronged intervention strategy, including **daily dollar sales** of approximately USD 1 billion in the spot market and the announcement of a **USD 5 billion USD/INR buy-sell swap auction**.

What are the RBI's Currency Intervention Tools?

- **Spot Market Dollar Sales:** The RBI's first line of defense is directly selling US **Dollars from its Foreign Exchange Reserves** into the **spot market**.
 - This injects **immediate dollar supply**, matching the excess demand from importers and arresting the **rupee's free fall**.
 - However, selling dollars sucks out rupee liquidity from the domestic banking system. If left unchecked, this **would spike domestic interest rates and hurt economic growth**.
 - To counter this, the RBI conducts "Sterilized Interventions." When it sells **dollars (draining rupees)**, it simultaneously conducts **Open Market Operations (OMOs)**, purchasing **government bonds** to pump rupees *back* into the system.
 - Conversely, during periods of massive capital inflows, to prevent excess liquidity from causing **inflation**, the RBI issues short-dated government securities called **Market Stabilisation Scheme (MSS) bonds** to mop up the surplus rupees.
- **Forex Swap Auctions:** The RBI increasingly uses "Buy/Sell" or "Sell/Buy" Forex swaps to manage both currency and domestic liquidity simultaneously without permanently altering its forex reserves.
 - **Buy/Sell Swap (To Inject Rupees):** The RBI buys US Dollars from banks and gives them Indian Rupees in return with a forward agreement to buy them back after a fixed tenor (e.g., 3 years). This instantly releases cash into the banking system.
 - **Sell/Buy Swap (To Absorb Rupees):** The RBI sells US Dollars to banks and takes back rupees.
 - This sucks excess cash out of the market to control inflation or currency volatility.
 - **May 2026 USD 5 Billion Swap:** With the rupee nearing over 95 against the USD due to the **US-Iran conflict**, the RBI announced a USD 5 billion buy-sell swap auction.
- **Contractionary Monetary Policy** : When forex interventions are not enough, the RBI uses monetary policy to make betting against the rupee unprofitable.
 - **Repo Rate Hikes:** The **Monetary Policy Committee (MPC)** will hike the benchmark **Repo Rate**.
 - Higher interest rates improve the yield differential with the US Federal Reserve, making **Indian debt more attractive to foreign investors, thereby encouraging capital inflows and supporting the rupee**.
 - **Liquidity Adjustment Facility (LAF) Squeeze:** When the currency is under severe speculative attack, the RBI can tighten the **Liquidity Adjustment Facility (LAF)** corridor, restricting banks' borrowing windows.
 - This temporarily pushes up the **overnight call money rate**. Because speculators usually **borrow rupees to buy dollars (shorting the INR)**, a higher cost of borrowing rupees makes this speculative trade prohibitively expensive, **forcing them to abandon their positions**.
- **Aggressive Pre-Market & NDF Interventions:**
 - **Breaking the Speculative Loop:** To prevent panic during **consecutive days of depreciation**, the RBI deploys "pre-market" interventions.
 - By heavily selling dollars through **state-run banks before the onshore spot market officially opens**, the RBI forces the rupee to **open stronger**, breaking the psychological negative feedback loop for traders.
 - **Curbing the Offshore Market (NDF):** Speculators often short the rupee in the offshore **Non-**

Deliverable Forward (NDF) markets (like London or Singapore).

- The RBI restricts domestic banks' net open forex positions and occasionally intervenes **directly in the NDF market to inflict losses on speculators betting against the rupee.**
- **Special Dollar Window:** Oil Marketing Companies (OMCs) are the largest buyers of US dollars in India.
 - Their daily dollar purchases to pay for crude oil imports can create massive sudden demand, crashing the rupee.
 - To prevent this, the RBI opens a **"Special Dollar Window" or dedicated foreign currency credit line** specifically for OMCs.
 - Instead of buying dollars from the **open spot market** (which spikes the dollar price), OMCs buy dollars directly from the **RBI or designated state-run banks (like SBI)** at a fixed swap rate. This was used effectively during the **2013 currency crisis**.
- **Capital Flow Management Measures (CFMs):** When capital leaves the country too fast, the RBI alters macroeconomic rules to slow down outflows and encourage immediate inflows.
 - **Easing ECB and FPI Norms:** The RBI temporarily relaxes the rules for **External Commercial Borrowings (ECBs)** (allowing Indian companies to borrow more dollars from abroad) and eases caps on **Foreign Portfolio Investors (FPIs)** investing in Indian government bonds.
 - This actively attracts foreign capital into the country.
 - **Promoting Masala Bonds:** Encourage Indian corporate borrowers to issue **Masala Bonds (rupee-denominated bonds issued in overseas markets).**
 - Because the bond is denominated in rupees, the foreign investor bears the **currency depreciation risk, not the Indian borrower.**
 - This brings foreign capital into the country without creating any future dollar repayment pressure on the Indian forex market.
 - **Capping Speculative Exposure:** In 2026, the RBI implemented a strict **CFM by capping banks' Net Open Position (NOP)** in foreign currencies at just USD 100 million per day.
 - This forced banks to unwind excess dollar holdings, increasing dollar supply in the forex market, reducing speculative pressure against the rupee, and helping stabilize the currency while curbing rapid depreciation.
- **Mobilizing NRI Capital (Unconventional Crisis Tools):** During severe Balance of Payments (BoP) crises, the RBI directly incentivizes **Non-Resident Indians (NRIs) to bring dollar liquidity into the country.**
 - **1998 Asian Financial Crisis:** Faced with global market contagion and post-nuclear test sanctions, the RBI supported the **issuance of Resurgent India Bonds (RIBs)** to overseas Indians, successfully raising around USD 4.2 billion to shore up reserves.
 - **2013 "Taper Tantrum":** When the US Federal Reserve signaled an end to its quantitative easing, foreign capital fled emerging markets.
 - The RBI opened a **special Foreign Currency Non-Resident (Bank) [FCNR(B)] deposit window.** The RBI offered domestic banks a highly **subsidized hedging rate**. This mobilized a massive USD 26 billion in weeks, instantly **stabilizing the currency.**

Frequently Asked Questions (FAQs)

1. What is RBI's first-line tool to defend the rupee?

The RBI usually sells US dollars in the spot market to increase dollar supply and reduce excessive pressure on the rupee.

2. What is sterilized intervention?

It is when the RBI offsets the liquidity impact of forex intervention through tools like Open Market Operations or MSS bonds.

3. How does a forex swap help the rupee?

A forex swap supplies rupee liquidity and manages volatility without permanently reducing foreign exchange reserves.

4. Why does RBI tighten the LAF during rupee stress?

It raises borrowing costs for speculators who short the rupee, making speculative bets more expensive and less attractive.

5. What are Capital Flow Management Measures?

These are temporary policy steps used to slow capital outflows or encourage inflows, such as easing ECB norms or capping speculative exposure.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/dmQPaouCyx0](https://www.youtube.com/embed/dmQPaouCyx0)]

UPSC Civil Services Examination Previous Year Question (PYQ)

Prelims

Q1. Which one of the following is not the most likely measure the Government/RBI takes to stop the slide of the Indian rupee? (2019)

- (a) Curbing imports of non-essential goods and promoting exports
- (b) Encouraging Indian borrowers to issue rupee-denominated Masala Bonds
- (c) Easing conditions relating to external commercial borrowing
- (d) Following an expansionary monetary policy

Ans: (d)

Q2. Consider the following statements: (2021)

1. The effect of devaluation of a currency is that it necessarily improves the competitiveness of the domestic exports in the foreign markets
2. increases the foreign value of domestic currency
3. improves the trade balance

Which of the above statements is/are correct?

- (a) 1 only
- (b) 1 and 2
- (c) 3 only
- (d) 2 and 3

Ans: (a)

Mains

Q. How would the recent phenomena of protectionism and currency manipulations in world trade affect macroeconomic stability of India? **(2018)**

Chandrayaan-3 Hop Experiment



Source: IE

Recently, Indian Space Research Organisation scientists published findings from **Chandrayaan-3's** Vikram lander "hop" experiment and Chandra's Surface Thermophysical Experiment, revealing a

layered structure within the Moon's upper regolith.

- **ChaSTE Experiment:** The **ChaSTE instrument**, equipped with temperature sensors and a heating probe, penetrated the regolith to study the Moon's thermal properties.
- **Hop Experiment:** The Vikram lander fired its engines and performed a hop of about 50 cm using residual propellant, helping scientists study surface behaviour at a different location.
- **Key Findings:** The experiment provided insights into **surface erosion, compaction, layered stratigraphy, geotechnical variability and lunar thermal profiles**.
- **Lunar Regolith:** Scientists clarified that the Moon's surface layer is more accurately termed "**lunar regolith**" rather than lunar soil, consisting of shattered rocks and jagged glass-like particles.
- **Layered Structure:** The study revealed a **two-layered cake-like structure** within the top few centimetres of the lunar surface.
- **Top Layer Characteristics:** The upper **2-6 cm layer** was found to be highly cohesive, sticky and hyper-porous, functioning like a **thermal blanket**. Scientists observed major density variation within a few centimetres, with bulk density increasing from **750 kg/m³ to 1600 kg/m³** below the surface.
 - This porous layer is significant for the **storage of subsurface water-ice molecules** and selection of sites for future lunar bases and habitats.
- **Significance:** These findings are important for understanding rocket plume-surface interaction, lunar surface stability, thermal behaviour, possible water-ice assessment and future lunar base planning.



Operation Navjeevan



Source: TH

Recently, 27 Maoists, including commanders and strike squad members of the banned CPI (Maoist) and Jharkhand Jan Mukti Parishad (JJMP), surrendered before Jharkhand Police in Ranchi under **Operation Navjeevan**.

- The operation was carried out by **Jharkhand Police** , its anti-Naxal unit Jharkhand Jaguar, and the **COBRA** of the **Central Reserve Police Force (CRPF)** .
- The surrender took place under the Jharkhand Government's **Surrender and Rehabilitation Policy** aimed at reintegrating former Maoists into society.
- This comes following the **Union Home Minister's declaration** that Maoists have been “**more or less wiped out**” from several key Left Wing Extremism-affected areas, and his assertion that the country is moving towards becoming “ **Naxal-free**” in line with the Government of India's target to eliminate **Left Wing Extremism by March 2026**.
- The Government has also adopted several measures to curb Left Wing Extremism, including the **SAMADHAN Doctrine** , **Operation Octopus**, **Operation Double Bull**, **Operation Kagar**, **RCPLWEA** , etc.

LEFT WING EXTREMISM

ABOUT.

- 🕒 **Originated:** 1967 uprising in Naxalbari, West Bengal
- 🕒 **Aim:** Societal and political change through revolutionary methods

IDEOLOGY.

- 🕒 Overthrow central government through armed revolution (Violence and guerilla warfare)
- 🕒 Establishment of a communist state based on Maoist principles

FACTORS RESPONSIBLE.

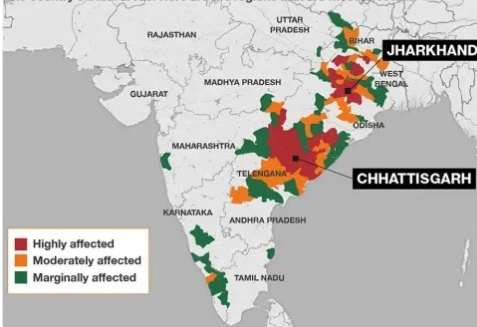
- 🕒 **Massive displacement of tribal population;** Due to development projects, mining operations
- 🕒 **Tribal dissatisfaction;** Forest (Conservation) Act 1980 prohibits tribes from harvesting forest resources
- 🕒 **Poverty and lack of sustainable means;** Driving factors for joining Naxalite movement
- 🕒 **Lack of effective governance;** Insufficient technical intelligence against Naxalism

STATES AFFECTED BY LWE.

- 🕒 **Red Corridor:** Experiences severe Naxalism–Maoist insurgency
- 🕒 Chhattisgarh, Jharkhand, Odisha, Bihar, West Bengal, Andhra Pradesh, Telangana, Maharashtra, Madhya Pradesh and Kerala

A map of India's Maoist conflict

A crackdown on Maoist rebels has led to a rise in the number of casualties in the country's tribal areas. Here are the regions that are most affected.



GOVERNMENT INITIATIVES TO CURB LWE.

- 🕒 National Policy and Action Plan to address Left Wing Extremism 2015
- 🕒 **SAMADHAN Doctrine**
 - 🕒 **S-** Smart Leadership
 - 🕒 **A-** Aggressive Strategy
 - 🕒 **M-** Motivation and Training
 - 🕒 **A-** Actionable Intelligence
 - 🕒 **D-** Dashboard Based KPIs (Key Performance Indicators) and KRAs (Key Result Areas)
- 🕒 **H-** Harnessing Technology
- 🕒 **A-** Action plan for each Theatre
- 🕒 **N-** No access to Financing
- 🕒 **Special Central Assistance (SCA)** in Public Infrastructure and Services
- 🕒 **Operation Green Hunt**
- 🕒 **Greyhounds** (Elite commando force in Andhra Pradesh)
- 🕒 **Bastariya Battalion** (local recruits in Chhattisgarh who know the language and terrain that could generate intelligence and conduct operations)

Countering Naxalism - Bandyopadhyay Committee (2006)

- Highlighted the lack of governance, economic, socio-political and cultural discrimination against the tribals
- Recommended tribal-friendly land acquisition and rehabilitation



Read more: [Eliminating Maoist Insurgency](#)

BSF's 1st All-Women Everest Expedition



Source: TOI

Recently, the **Border Security Force (BSF)**'s **first all-women mountaineering expedition** successfully scaled **Mount Everest**, creating a landmark achievement in India's adventure and security history.

- **Mission Vande Mataram:** The expedition was named “**Mission Vande Mataram**” as part of the **BSF’s Diamond Jubilee celebrations** and the **150 th anniversary of the national song ‘Vande Mataram’** .
 - Under the “**Clean Himalaya - Clean Glaciers**” campaign, the team also collected waste from Everest camps for proper disposal.
- **Border Security Force (BSF) :** The **Border Security Force (BSF)** was established on 1 st December 1965 after the India-Pakistan war and is regarded as India’s “**First Line of Defence**”, guarding borders with Pakistan and Bangladesh.
 - It is one of the **seven Central Armed Police Forces (CAPFs)** of India and functions under the **Ministry of Home Affairs (MHA)** .
- **Mount Everest:** Mount Everest is the world’s highest mountain peak with a height of 8,849 metres (29,032 feet), located in the **Himalayas** on the Nepal-China border.
 - In Nepal, it is known as **Sagarmatha** , while in Tibet it is called **Chomolungma** .
 - It was first successfully climbed in **1953 by Sir Edmund Hillary of New Zealand and Tenzing Norgay Sherpa of Nepal** .



Read more: [Extending BSF Jurisdiction](#)